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Automatic 3-dimensional cephalometric landmarking based on active shape models in related projections

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Introduction: This article presents a novel technique for automatic cephalometric landmark localization on 3-dimensional (3D) cone-beam computed tomography (CBCT) volumes by using an active shape model to search for landmarks in related projections. **Methods:** Twenty-four random CBCT scans from a public data set were imported and processed into Matlab (MathWorks, Natick, Mass). Orthogonal coronal and sagittal projections (digitally reconstructed radiographs) were created, and 2 trained active shape models were used to locate cephalometric landmarks on each. Finally, by relating projections, 18 tridimensional landmarks were located on CBCT volume representations. **Results:** From our 3D gold standard, a 3.64-mm mean error in localization of cephalometric landmarks was achieved with this method, with the highest localization errors in the porion and sella regions because of the low volume definition. **Conclusions:** The proposed algorithm for automatic 3D landmarking on CBCT volumes seems to be useful for 3D cephalometric analysis. This study shows that a fast 2-dimensional landmark search can be useful for 3D localization, which could save computational time compared with a full-volume analysis. Also, this research confirms that by using CBCT for cephalometry, there are no distortion projections, and full structure information of a virtual patient is manageable in a personal computer. (Am J Orthod Dentofacial Orthop 2018;153:449-58)

As is well known in orthodontics, cephalometry describes the morphology of the craniofacial skeleton and the skull measurements from cephalograms.[1] Cephalograms and digital radiographs are commonly used in conventional cephalometry, but they only provide information on a plane (coronal, sagittal, or axial) from the 3-dimensional (3D) space, unlike cone-beam computed tomography (CBCT) that provides high-resolution images without overlapping or distortion, which results in high-quality diagnostic images. Recently, CBCT was introduced to the dental community as a diagnostic tool and has became a standard imaging technique in orthodontics.[1] This is because tomography scans provide accurate 3D information about the patient's size and position.

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Recent and relevant studies for automated 3D cephalometric landmarking on CBCT volumes used volume analysis instead of surface analysis, where located landmarks are directly annotated in volume voxels. For this research, 4 state-of-the-art studies were identified. The first study by Gupta et al[2] proposed an algorithm to search for 20 cephalometric landmarks in 30 CBCT preprocessed images by grouping search sections of the head. They reported an average accuracy of 2.01 mm, with 64.67% of the landmarks in a range of 0 to 2 mm, 82.67% from 0 to 3 mm, and 90.33% from 0 to 4 mm. Shahidi et al[3] evaluated the design of a software to locate cephalometric landmarks in CBCT. They reported a mean error in 14 landmark localizations under 4 mm, and 63.57% of landmarks had a mean error less than 3 mm compared with manual localization (gold standard). Makram and Kamel[4] proposed a system for automatic localization of 20 hard tissue cephalometric landmarks using reeb graphs in 3D patient meshes where some nodes were considered as cephalometric landmarks. Ninety percent of their landmarks had a localization error less than 2 mm. Codari et al[5] presented a method to automatically locate cephalometric points from volumetric reconstructions. After automatic segmentation of the hard tissue, a nonrigid holistic register

Table I. Cephalometric landmark definitions

Landmark	*Definition*
Sella (S)	Midpoint of rim between anterior clinoid process in median plane
Nasion (N)	Midsagittal point at junction of frontal and nasal bones at nasofrontal suture
Basion (Ba)	Most inferior point on anterior margin of foramen magnum at base of clivus
Orbitale (O)	Most inferior point on infraorbital rim, right (OR) and left (OL)
Anterior nasal spine (ANS)	Most anterior limit of floor of nose at tip of ANS
Posterior nasal spine (PNS)	Point along palate immediately inferior to pterygomaxillary fossa
A-point (subspinale)	Most concave point of anterior maxilla
B-point (supramentale)	Most concave point on mandibular symphysis
Gonion (Go)	A point in the middle of the curvature at the left (GoL) and right (GoR) angles of the mandible
Pogonion (Pg)	Most anterior point along curvature of chin
Menton (M)	Most inferior point along curvature of chin
Porion (Po)	Most superior point of anatomic external auditory meatus, right (PoR) and left (PoL)
Gnathion (Gn)	Located perpendicular on madibular symphysis midway between Pg and M
Incisor inferior (Ii)	Incisal edge of the most prominent mandibular incisor
Incisor superior (Is)	Incisal edge of the most prominent maxillary incisor

was carried out between a reference template volume containing annotated cephalometric points and a study volume on which the points were located. These authors reported an average localization error of 1.99 mm on 21 points in 18 CBCT volumes.

CBCT volumes allows reforming the 3D structure of the skull; then 2-dimensional (2D) conventional simulated x-ray images, or digitally reconstructed radiographs (DRRs), can also be calculated for conventional cephalometry. Therefore, DRRs can be defined as 2D simulated approximations of a radiograph with the advantages of maintaining the patient's size and position.[6] Then cephalometry can be performed by using more than 1 x-ray image; eg, Moshiri et al[7] presented a cephalometry method using DRRs obtained from CBCT. They compared the accuracy of linear measurements made on conventional digital radiographs and DRRs, concluding that there is no advantage in using conventional radiographs over DRRs for cephalometry.

Table II. Step-by-step 3D approach to CBCT cephalometric landmarking on hard tissues

Start	*Loading DICOM data into 3D viewer*	*Automated*
Step 1	Computing coronal and sagittal DRR projections	Automated
Step 2	Initializing ASM search by clicking close to sella	Manual
Step 3	Coronal and sagittal planes and landmark correlations	Automated
End	Definition of 3D cephalometric landmarks in CBCT volume	Automated

Cephalometric landmarks have been traditionally studied in 2 dimensions for cephalometry in orthodontics. However, since the head is a 3D structure, it is necessary to locate its position in 3D space to be used in orthodontics. In this study, we explored statistical shape modeling for automatic 3D cephalometric landmarking by relating DRRs using an active shape model (ASM).[8] An ASM is a simple and robust tool for statistical shape analysis. We describe how our ASM is constructed and used for automatic 3D cephalometric landmarking by relating 2 projections.

MATERIAL AND METHODS

The sample for this experiment consisted of 24 CBCT head volume scans. No demographic data were available, and the images were not identified by age, sex, or ethnicity. The CBCT images were randomly selected from a public data set: the Virtual Skeleton Database[9] from the Swiss Institute for Computer Assisted Surgery Medical Image Repository. As shown in Table I, 18 cephalometric landmarks were identified by consensus on the sagittal and coronal projections of each volume to train an ASM. To establish the true positions of the selected cephalometric landmarks, manual annotation was independently made twice by 2 observers with varying landmarking experience. From 1 researcher, intraoperator data were obtained by placing 10 times the 18 landmarks on 24 CBCT scans. Means and standard deviations were calculated from the manual annotations for each landmark, and the mean was taken as the anatomic gold standard. To set the 3D ground truth in volumes, multiplanar reconstruction was rendered in 3DSlicer,[10] and landmarks were saved as a list of fiducial points.

Volumes were provided as DICOM image sets consisting of approximately 320 slices with 0.4-mm isometric voxels. Since the volume data stores density values of the scanned body material in voxels, data were loaded into Matlab (MathWorks, Natick, Mass)

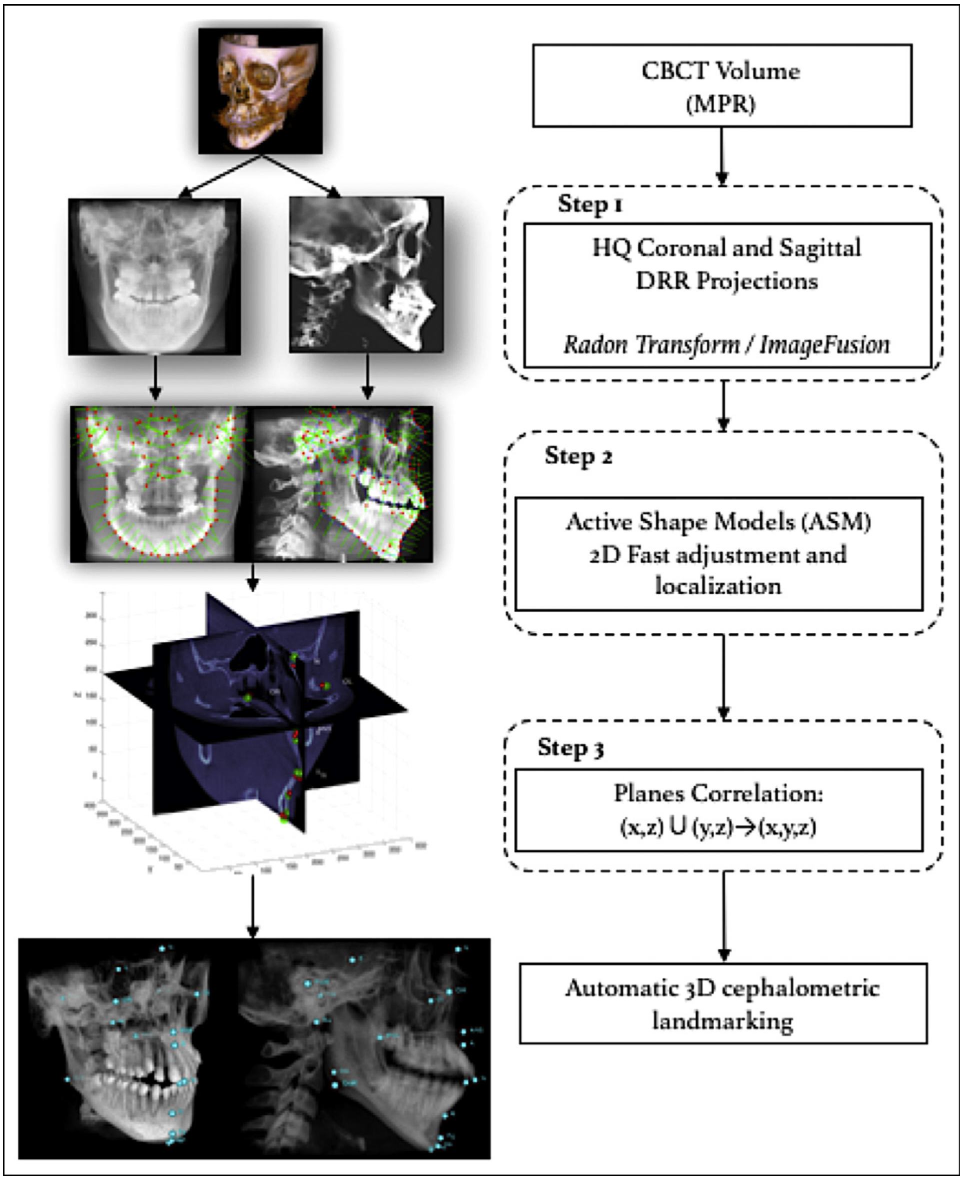

Fig 1. General scheme of our technique for the automatic landmark localization in CBCT volumes.

without preprocessing to synthetize various types of intensity projections for CBCT volume data, also known as DRRs.

The proposed technique for automatic 3D landmarking in CBCT consists of 3 main steps described in Table II and illustrated in Figure 1. After loading the DICOM data, the first step is to compute the corresponding DRR in coronal and sagittal projections for each volume. In the second step, the ASMs are initialized by manual 2D localization of sella in the sagittal DRR. In this step, ASMs are adjusted into the bone shapes in the sagittal and coronal DRRs; then 18 landmarks are located in separate perpendicular projections. In the third step, the coronal and sagittal projections are related, in which landmarks located in both projections are used to calculate a 3D point coordinate for each landmark. Finally, 3D landmarks can be visualized on the multiplanar reconstruction representation of the volume. Additionally, Frankfort horizontal, sella-nasion, facial, mandibular, palatal, and basion-nasion planes can also be calculated and shown in the volume.[11]

The principle of DRRs as intensity projections is to project all slices within a volume into a single 2D image, directly based on the attenuation absorption law.[12]

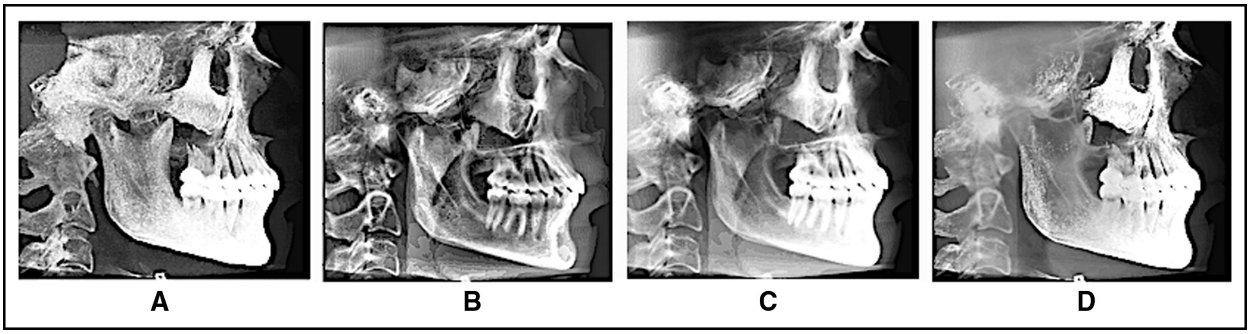

Fig 2. Samples of **A,** maximum intensity projection; **B,** mean intensity projection; **C,** standard deviation intensity projection; **D,** resulting image by image fusion used to annotate the training sets for the ASM. Coronal images were also calculated.

$$I = I_0 \sum_{i=1}^{n} e^{-\mu_i d_i}$$

where I_0, is the original ray value, i is the voxel through which a ray passes, μ is the linear attenuation coefficient of the material in voxel i, and d_i is a segment between the entrance and output points of the ray in voxel i.

Each pixel of a 2D final image is a combination of all voxels with the same 2D coordinates in every projected image. The way these voxel values are combined is determined by the selected algorithm. We have analyzed projection algorithms that give high contrast and good definition of structures for cephalometric landmark locations. Conventional lateral cephalograms and even DRRs generally suffer from low contrast and visibility of many structures. In this study, we used an effective enhancement method for DRRs by image fusion of different orthogonal projections.[13]

The CBCT slices and their thicknesses were extracted from DICOM meta data images using Matlab. The CBCT volumes we used have a 0.4-mm isotropic voxel resolution. Volumetric reconstruction occurred in real time and provided contiguous color-related perpendicular axial, coronal, and sagittal 2D multiplanar reconstruction slices. Maximum intensity projection and standard deviation intensity projection were calculated and fused under image fusion parameters to give enhanced and natural-looking contrast images without artifacts. This orthogonal DRR projection maintains the real size in measurements by cephalometric landmarks and produces matches to actual CBCT volumes with 0% magnification in comparison with 5% to 8% magnification in conventional perspective radiographs.[14,15] Figure 2 shows image fusion projections to highlight bone structures. Finally, these images were used to train an ASM.

For this study, the ASM uses statistical deformations of shapes from a training set of hand-annotated images to create a model that will be fitted to unseen x-ray images (conventional or DRR) for automatic cephalometric landmarking.[8] The process implies the manual placement of points in each radiograph following an established path to ensure correspondence in all images. Figure 3 shows the alignment of every annotated projection in the same space, whereas Figure 4, *A* and *B*, shows our template that contains cephalometric landmarks and pseudo landmarks spaced along strong edges that regularly appear on lateral and frontal head radiographs. In total, 70 points were used to define a shape for coronal projections and 95 points to define a shape for sagittal projections. Training images were imported into a created manual cephalometric landmarking program in Matlab, and the landmarks were identified by using a cursor-driven pointer. A custom analysis in our program was developed to assist the observer in identifying and annotating a set of specific structure points and cephalometric landmarks on the images to build our models (Table I).

Head structures used to locate cephalometric landmarks are described by n landmark points (manually located), determined in our set of 24 coronal DRR and 24 sagittal DRR training images. Landmark points $(x_1, y_1), \ldots, (x_n, y_n)$ are grouped as follows.

$$x = (x_1, y_1, \ldots, x_n, y_n)^T$$

Annotated shapes were mutually aligned by applying the Procrustes algorithm, following by principal components analysis to compute mean shapes, eigenvalues, and eigenvectors from which an ASM is composed using variation modes. All modes imply complex variations that are represented by principal components. With Procrustes analysis alignment, we could reduce the effects of external transformations to better represent underlying internal shape variations.

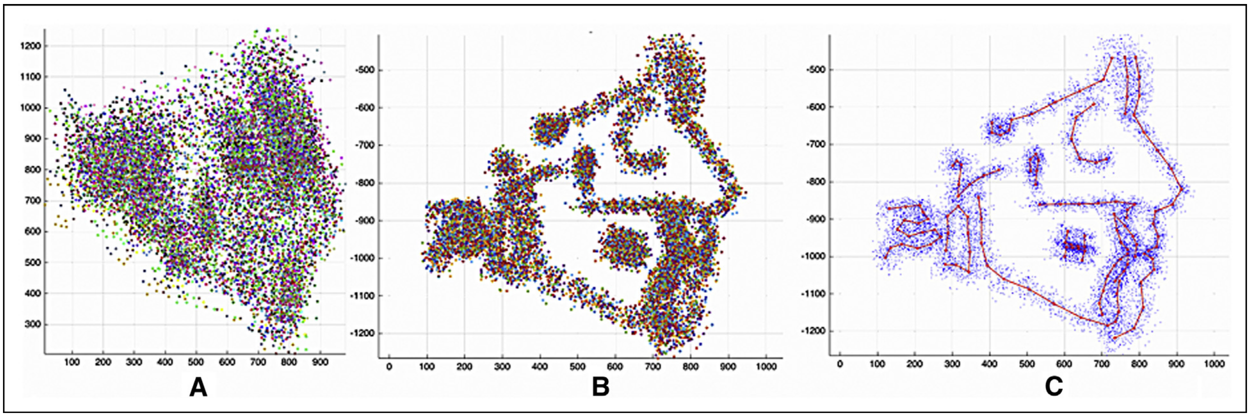

Fig 3. Initial process for shape alignment of manual annotations for variation modes calculation using principle components analysis: **A,** all annotated shapes before alignment; **B,** annotated shapes aligned by Procrustes analysis; **C,** mean shape in sagittal views.

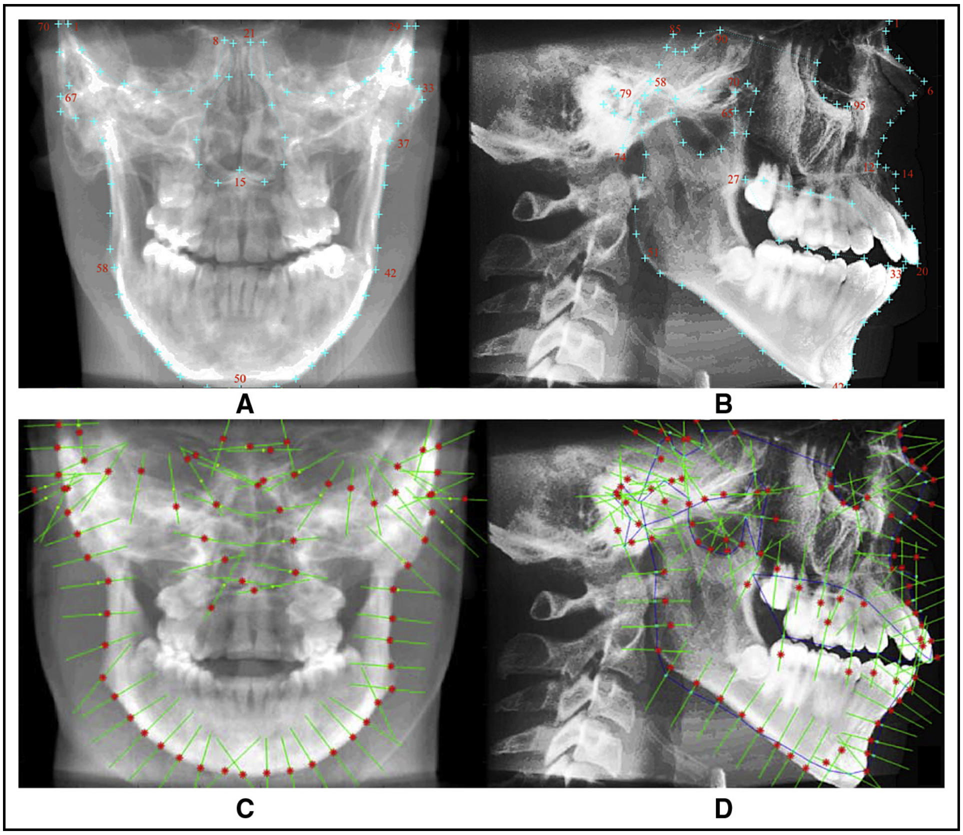

Fig 4. Manual annotation in **A,** coronal and **B,** sagittal projections and **C,** ASM coronal and **D,** sagittal fitting of the landmarking model. *Green lines* in *C* and *D* show gray profiles; *red dots* from the actual position of the model are the pull vectors to set points in high-contrast bone structures. Note that we are not using soft tissues in this adjustment.

A local gray intensity model was also computed around each point, which describes the typical image structure around each landmark obtained from pixel profiles. We used the Bresenham algorithm[16] to compute pixel profiles around each landmark by calculating curvature lines in all contours.

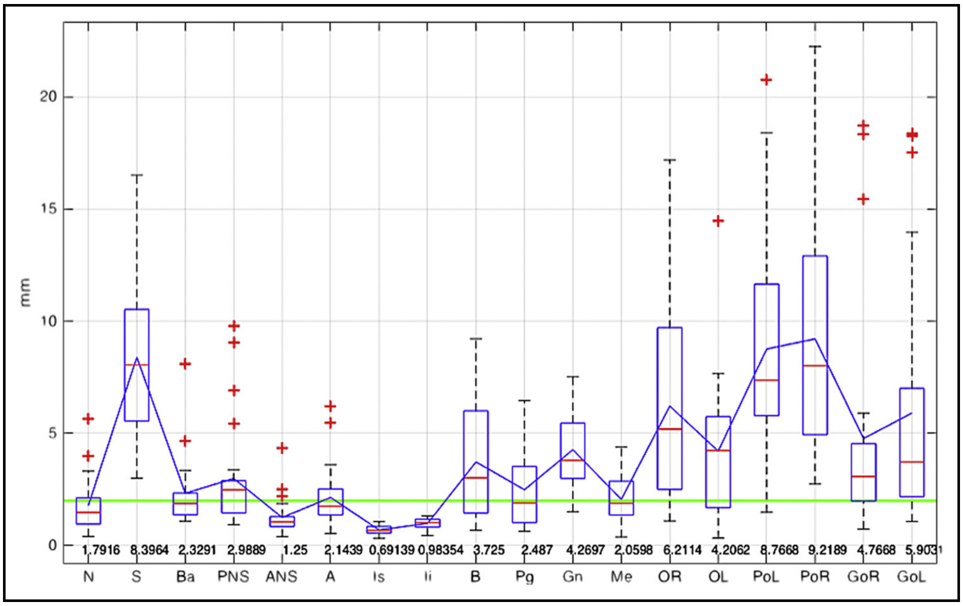

Fig 5. Localization errors box plot for the 18 landmarks, which in essence is the Euclidean distance between the ground truth and the estimated location by the ASM search in related planes.

Using those normalized derivative profiles $g_1, \ldots, g_s$, the mean profile $\bar{g}$ and the covariance matrix S_g are computed for each landmark. This allowed us to compute the Mahalanobis distance between a new profile g_i and the profile model.

$$f(g_i) = (g_i - \bar{g})^T S_g^{-1} (g_i - \bar{g})$$

Hence, minimizing the Mahalanobis distance[17] $f(g_i)$ is equivalent to maximizing the probability that g_i is in the distribution $\{g_1, \ldots, g_s\}$.

As we can see in Figure 4, both models (gray intensity and shape) were used to automatically fit the mean shape to unseen images. Such unseen images were DRRs created from CBCT images, yielding a location on the image for each cephalometric landmark.

All new shapes are fitted in an iterative way, starting from the mean shape. Each landmark in the model is moved n pixels along a perpendicular direction on the contour on either side. Evaluating $2n + 1$ pixels, we used $n = 35$ in this experiment. Each model's landmark is put at the position with the lowest Mahalanobis distance between g_i and $\bar{g}$ profiles in 5 pixel steps. After moving all landmarks, the shape model is fitted to the displaced points, and the actual model is updated. This is repeated n_{max} times until a 1-pixel movement is reached. An adjustment to a DRR is shown in Figure 4, *C* and *D*. An ASM test by the leaving-one-out algorithm with 24 models was performed to assess the accuracy with unseen projections. Figure 5 shows the results for each automatically located landmark in related planes by measuring the Euclidean distance between the ground truth and the ASM automatic detection. Localization errors between 0.5 and 6 mm were obtained; therefore, improved adjustments were needed specifically for sella and porion. A new training for the gray intensity model by k-means bone segmentation was done before image fusion to increase the DRR imaging quality, demonstrating better performance in ASM adjustment.[15]

From the ASM search described above, we obtained approximated cephalometric point locations in the sagittal plane, πs, and in the coronal plane, πc. Since πs and πc geometric objects have no volume—ie, are 2D—coplanar with the Cartesian axes $(y,\ z)$ and $(x,\ z)$, respectively; each contains a finite number of cephalometric landmarks that were related for the sagittal plane as $P_{\pi s} = (0, y_i,\ z_i)$ and for the coronal plane as $P_{\pi C} = (x_i, 0,\ z_i)$.

This step was aimed to find a 3D point with components corresponding to Πs and Πc to represent every cephalometric landmark as an 18 3D points set P. In this way, P is our approach approximation of the location of 3D cephalometric landmarks associated with a CBCT volume. Figure 6 shows the related coronal and sagittal projections after ASM adjustment.

RESULTS

The ASM algorithm for automatic medical image segmentation was used for automatic cephalometric landmark localization. It was applied to DRR projections, and ASM fitting planes were related to 3D CBCT volumes.

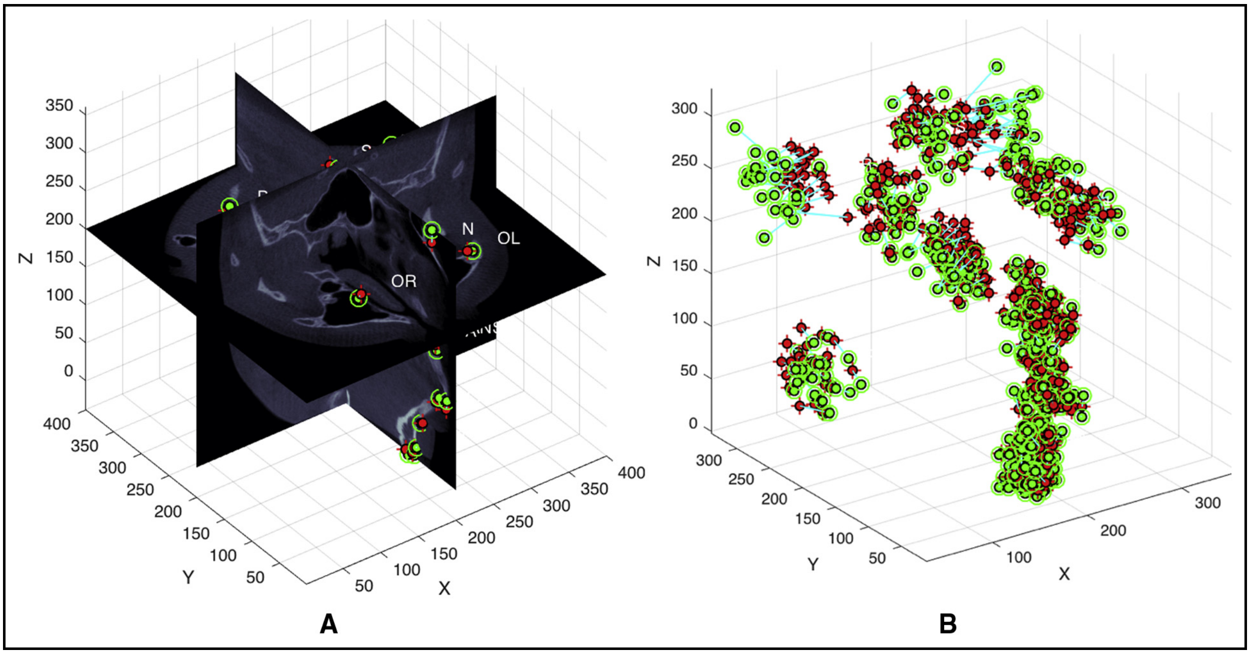

Fig 6. Final localization results: **A,** projections relation, after ASM adjustment in coronal and sagittal projections from which 18 cephalometric landmark coordinates were obtained; **B,** estimated cephalometric landmarks (*red*) and respective ground truth (*green*).

For this study, our localization method was fully implemented in Matlab using 3DSlicer visualization when needed. The leave-1-out algorithm to evaluate the model trained by 70 points (cephalometric and pseudo landmarks) in 24 DRR coronal projection images and 95 points in 24 DRR sagittal projection images was implemented to assess localization accuracy. The reproducibility of 18 hard tissue landmarks from the set was evaluated on DRRs and CBCT volumes. From the intraoperator data, 12 of the 18 landmarks were found to be reproducible within a 1.0-mm standard deviation.

Figure 7 shows mean localization errors for each landmark that are referenced by a 2-mm line that represents the maximum acceptable error and our mean global error line. The localized cephalometric landmarks were annotated on their volumes; different visualizations are illustrated in Figure 8. In Table III, the measured mean error dimension from each localized cephalometric landmark is listed. The automatically located landmarks were compared with those obtained from the 3D ground truth annotated in the CBCT volumes. As we can see, the location mean error for the landmarks was 3.64 mm. We believe that the largest localization men errors, porions and sella landmarks, are due to low definition of these regions in our data set. Furthermore, porion is well known as a difficult point to locate.[18]

DISCUSSION

Although today there are numerous techniques and softwares designed to assist orthodontists in cephalometric analysis, there is no tool to directly perform the automatic landmarking process for 3D cephalometry in volumes as proposed. Recently, 3D technology is increasing in hospitals and research settings; unfortunately, some care sectors are still using standard photographic and radiographic views (2D) as the leading imaging diagnostic tool for orthodontic care.[19,20] Lateral cephalograms and photographs, frequently taken from different hardwares, do not have comparable degrees of magnification[6,21]; then it is difficult to standardize the use of these images. Thus, the use of DRRs may be a standard imaging tool without conventional magnification using CBCT for 3D cephalometry. It has interesting advantages for the future in orthodontics, such as reduced radiation exposure,[22] no occlusion, and real and natural shape of patients' craniofacial tissues. The main advantage we identified for using these 3D images is that we can perform automatic computed analysis, such as cephalometry, by voxel intensity and image shape processing. On the other hand, CBCT acquisition systems may produce different Hounsfield unit density values (radiographic densities) among different areas of the scanned volume in the same structures. For example, in the mandible, there are specific gray level values in lower areas and significantly

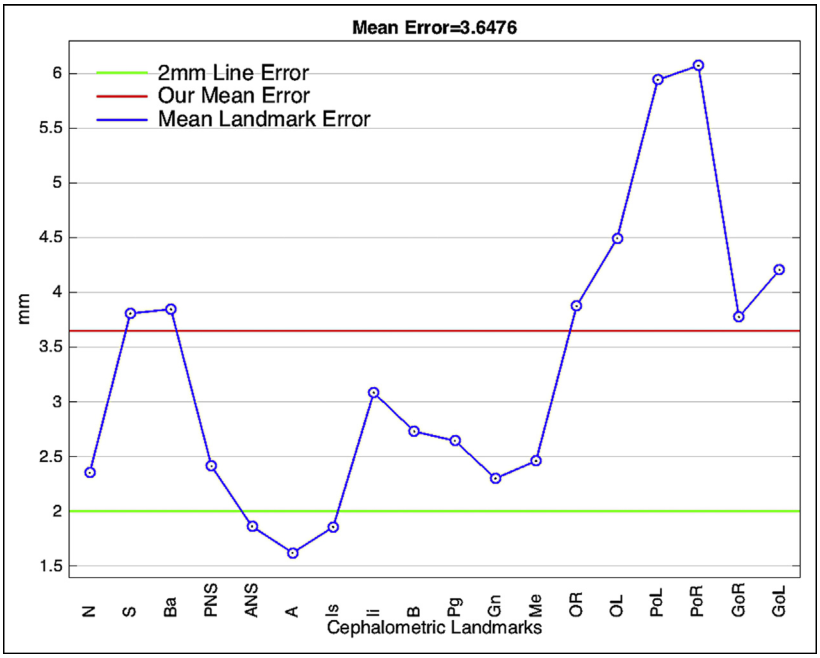

Fig 7. Landmark mean localization errors.

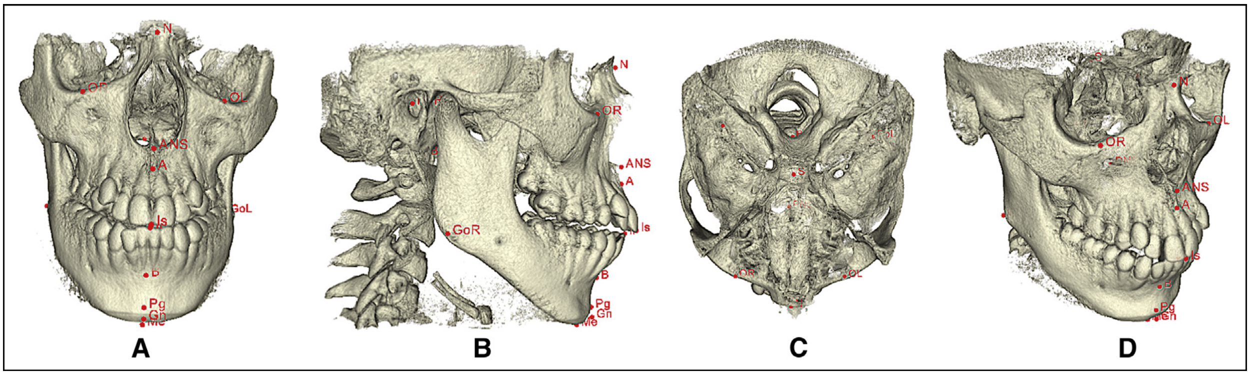

Fig 8. Three-dimensional automatic landmarking in a CBCT volume: **A,** coronal; **B,** sagittal; **C,** axial; **D,** isometric views of our related projections approach.

different values in upper areas.[23] Then, for automatic analysis such as cephalometric landmarking, this image processing could be prone to errors. Additionally, CBCT images produce DRR orthogonal projections with no differential magnification in bilateral or midsagittal structures. We believe that our mean accuracy of 3D landmark localization in CBCT images is due to no homogeneous data compared with conventional lateral cephalometric training images. Errors in localization are also due to head malpositioning,[22,24] radiographic method, and intraobserver and interobserver errors in landmark annotation.

Within the results of this study, intraobserver and interobserver accuracies in landmarking must be studied, and it is suggested that important cephalometric landmarks such as porion showing poor reproducibility for both interoperator and intraoperator data should be used with caution by an expert orthodontist. However, we believe that our approach could save time and improve reproducibility in some orthodontic practices: eg, for definition of both skeletal and facial types, evaluation of symmetry, relationships between the maxillary and mandibular basal bones, and dental relationships, and also to facilitate selection of a treatment plan.

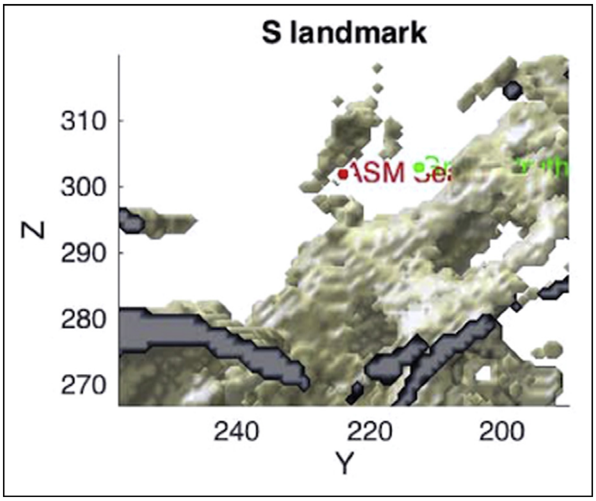

Fig 9. Results of the adjustment in a cropped volume of sella in the sagittal view; due to the low definition in this region, inherent to the cone-capturing device, localization is prone to greater errors. The same problem happens in porion regions.

Table III. Landmark detection rates with mean errors and standard deviations

Landmark	*Mean error (mm)*	*SD*
N	2.3509	0.9993
S	3.8061	1.7204
Ba	3.8448	1.4577
PNS	2.4127	1.066
ANS	1.8615	0.8325
ANS	1.6205	0.6956
Is	1.8561	0.6343
Ii	3.083	0.6514
B	2.7293	0.506
Pg	2.6435	0.9646
Gn	2.3001	0.8792
Me	2.4604	1.0211
OR	3.8735	1.7871
OL	4.8932	2.2626
PoL	9.7175	3.8384
PoR	8.225	3.1761
GoR	3.7738	1.4079
GoL	4.2042	1.8479
Mean	3.6476	1.43045

CONCLUSIONS

This article describes a novel technique for automatic 3D landmarking using related DRR projections toward a complete 3D automatic cephalometry. Coronal and sagittal projections were used to train 2 ASMs. Mean accuracy of 3.64 mm was achieved for the 18 landmarks in our CBCT images. Sella and porion were difficult to locate because of their irregular shapes, positional variations, and low definitions of those regions in the DICOM volumes (see Fig 9). The weakness of the proposed technique is the high error in sella and porion; subsequent studies will be needed to improve this. For now, the operator may need to perform manual adjustments in porion and sella after final automated landmark identification.

The proposed algorithm for automatic 3D landmarking on CBCT volumes may be useful for 3D cephalometric analysis. This study shows that a fast 2D landmark search can be useful for 3D localization, which could save computational time compared with a full-volume analysis. Furthermore, this research confirms that using CBCT for cephalometry, no-distortion projections, and full structure information of a virtual patient can be managed with a personal computer.

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